Summary & Teaser

- Key concepts learned:
  - Couplings, bistochastic matrices, Birkhoff, Kantorovich.
- Teaser: This formulation gives us one more thing ….

The Wasserstein Distance

When $c(x, y) = d(x, y)^p, p \geq 1$, Kantorovich OT is called **Wasserstein Distance**

$$\mathrm{W}_p(\alpha, \beta) = \left(\inf_{\pi \in \Pi(\alpha,\beta)} \int_X c(x, y)^p \, \mathrm{d}\pi(x, y) \right)^{1/p}$$